Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.0000.DOI

Does an Asymmetric-Reward E-DDQN Improve Safety-Prioritized Cybersickness Detection in Virtual Reality? A Leakage-Controlled Comparison with Standard DQN and Cost-Sensitive Baselines

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ABSTRACT We ask whether E-DDQN, a dueling Double Deep Q-Network whose reward penalizes a missed cybersickness event five times more than a false alarm, improves cybersickness detection in virtual reality. On logged data the task is a contextual bandit, and the 5:1 reward induces exactly the 5:1 cost-sensitive Bayes rule. On the public UFFCSData dataset (35 sessions, 9,390 one-second samples), row-level splits give ROC-AUC 0.992–0.995 versus 0.523 with session-grouped splits. Under a nested, session-grouped protocol fixed in advance, the 5:1 cost (mean over seeds; seed ensemble in parentheses) was 0.663 (0.726) for E-DDQN, 0.584 (0.536) for a standard DQN and 0.491 (0.485) for a cost-weighted random forest, with ROC-AUC 0.538, 0.661 and 0.688. E-DDQN's higher seed-ensemble cost than the random forest (+0.242, 95 % CI +0.052 to +0.528) was significant under the primary group bootstrap but not under an exact sign-flip sensitivity test (Holm-adjusted $p = 0.006$ and 0.083). In two further pre-specified search draws, E-DDQN never had a significantly lower cost than the standard DQN; its within-fold ROC-AUC ranged from 0.539 to 0.714 across draws, suggesting that search-draw sensitivity contributes to its weak primary result. A post-hoc logistic regression on elapsed time alone reached ROC-AUC 0.704; no primary model ranked seconds significantly better. Cost-optimal thresholds raised alarms on 82–97 % of non-event seconds. Bounded feature-space perturbations (a stress test, not an attack evaluation) changed ROC-AUC by at most 0.089. Asymmetric rewards encode a cost-sensitive operating point, not a new learning objective; reinforcement-learning detectors should be compared with cost-weighted and exposure-time baselines over several search draws.

INDEX TERMS Cybersickness, virtual reality, deep reinforcement learning, double deep Q-network, contextual bandit, cost-sensitive learning, data leakage, safety-critical detection.

I. INTRODUCTION

Cybersickness—nausea, disorientation and oculomotor strain caused by the sensory conflict of virtual reality (VR)—limits the use of head-mounted displays in rehabilitation, training and assessment [1]. It can also be induced deliberately by manipulating frame pacing, camera motion or field of view [2], [3], and attacks on a detector's inputs can suppress a mitigation [4]. A detector that recognizes cybersickness during immersion is therefore a safety function with unequal error costs: a missed event leaves the user exposed, whereas a false alarm causes a brief interruption.

Detectors have moved from post-exposure questionnaires to deep models of physiological, eye-tracking and gameplay data [5]–[10]. To encode unequal costs, several authors

cast classification as a sequential decision problem for reinforcement learning (RL) and trained deep Q-networks (DQNs) with class-dependent rewards [11], [12]. E-DDQN follows this line: a dueling Double DQN [13], [14] whose reward penalizes a missed event five times more than a false alarm, trained with class-balanced replay and soft target updates. Unlike [11], [12], it does not end an episode when an event is missed.

Two questions must be answered before such a detector can be credited with a safety benefit. Evaluation: gameplay telemetry is logged about once per second and self-reported discomfort changes rarely within a session, so a split that places seconds of one session in both training and test data rewards recognition of the session [15]; published results on the dataset used here come from a 70/30 split for which no session separation is reported [16], and an earlier, unpublished version of this study reported 96.88 % accuracy for E-DDQN after oversampling before splitting. Formulation: what an asymmetric reward adds to cost-sensitive learning [17] is rarely made explicit, and RL detectors are seldom compared with cost-weighted models tuned and thresholded under the same rules.

On the public UFFCSData dataset [18] we ask:

RQ1 What does the asymmetric reward add, formally, to cost-sensitive classification?

RQ2 Under leakage-controlled, session-grouped evaluation, does E-DDQN outperform a standard DQN, cost-weighted supervised models and an exposure-time baseline, and does any model meet a safety-prioritized operating target?

RQ3 As a secondary sanity check, how sensitive are the detectors to bounded manipulation of the telemetry channels? The dataset contains no manipulated sessions, so this is a feature-space stress test, not an evaluation of attacks.

The contributions are:

1) Formulation. Applying known results [17], [19], we show that on logged data the alarm cannot change future observations, so detection is a contextual bandit and the 5:1 reward induces exactly the Bayes rule $P(y = 1 \mid s) \geq 1/6$ of 5:1 cost-sensitive classification; class-balanced replay shifts the operating point actually learned (Section III-D).
2) A matched, leakage-controlled benchmark. E-DDQN, a standard DQN, a contextual-bandit variant and four cost-weighted supervised models are compared under one nested, session-linkage-grouped protocol fixed before testing, with the same search procedure, number of configurations (not coverage: the search spaces differ greatly in size), checkpoint and threshold rules (Figs. 1–3); two exposure-time baselines were added post hoc. Row-level splits of the same data give ROC-AUC 0.992–0.995, against 0.523 under session-grouped splits (Section IV-A).
3) Findings. E-DDQN did not have a significantly lower 5:1 cost than the standard DQN with its primary or two further, pre-specified search draws. Its seed-ensemble cost was higher than that of a cost-weighted random forest, significantly under the primary group bootstrap but not under an exact sign-flip sensitivity test, so this difference is not firmly established. No primary model ranked seconds significantly better than elapsed time alone, and no model combined high recall with a controlled false-alarm rate (Section IV).
4) Diagnosis. Across three search draws E-DDQN's within-fold ROC-AUC ranged from 0.539 to 0.714, which suggests that sensitivity to the search draw contributes substantially to its weak primary result; the draw, feature sets and other hyper-parameters varied together, so no single cause is isolated (Section IV-E).
5) Reproducibility. Code, the fixed protocol, per-sample predictions and executable checks of the splits and the replay buffer are released.

Section II reviews related work, Section III presents the methods and setup, Section IV the results and Section V the discussion; Section VI concludes.

II. RELATED WORK

A. CYBERSICKNESS DETECTION AND ITS EVALUATION

Questionnaires such as the Simulator Sickness Questionnaire [20] measure discomfort after exposure and cannot capture onset during immersion [1]. Sensor-based detectors use physiological signals [5], [6], eye and head tracking [7], EEG [8], consumer-grade physiological data with generative augmentation [9] and multimodal transformers [10]; explainable models identify dominant features and shrink detectors for deployment [16], [21], [22]. A systematic review notes that evaluation protocols vary widely [23]. The gameplay data used here were collected by Porcino et al. [18], [24], who related discomfort to rotation and acceleration [25]; TruVR reported 94.10 % accuracy on them with a 70/30 split [16], and they were used in work on attacks against detectors [4]. No session-level separation is reported for those splits. With 1-Hz samples and slowly changing labels, shared sessions let a model recognize sessions it has seen [15], [26], [27]; we quantify this on the gameplay data (Section IV-A) but do not re-evaluate the cited studies, whose pipelines are not available to us.

B. THREATS TO VR USERS AND DETECTORS

VR mediates perception and balance and therefore has a distinct attack surface [28], [29]: perceptual manipulations can be repurposed to cause harm [2], [30], frame-rate manipulation can induce sickness [3], and optimization-based input perturbations can stop deep detectors from triggering a mitigation [4], [31]. Our stress tests instead apply bounded, physically motivated changes to logged telemetry (Section III-H); neither approach measures what a manipulation does to a participant.

C. RL FOR IMBALANCED AND COST-SENSITIVE DETECTION

Double DQN reduces the over-estimation of DQN [13], [32] and dueling heads separate state value from action advantage [14]. Lin et al. [11] and Yang et al. [12] cast imbalanced classification as a DQN problem with larger minority-class rewards and end an episode when a minority sample is misclassified. Safe-RL work shows that penalties shape risk aversion [33], but the relation of asymmetric rewards to cost-sensitive thresholding [17], and the equivalence of a full-information contextual bandit with cost-sensitive classification [19], are rarely made explicit for RL detectors.

Gap. A safety benefit can be attributed to a detector design only if the evaluation controls leakage, the comparison with cost-weighted and minimal baselines is matched, and the decision rule is stated in terms of the costs it encodes. Supplementary Table S5 (Sec. S9.2) maps each gap to the cited evidence and to this study's response.

III. METHODS AND EXPERIMENTAL SETUP

Fig. 1 shows the implemented workflow. Stages 1–4 construct the data and the partitions without learning any statistic; stages 5–8 use only inner-training and inner-validation groups; each outer-test fold enters only stages 9–10, after every model choice for that fold is frozen.

A. DATA AND TARGET (STAGES 1–2)

UFFCSData [18], [24], [25] was recorded while volunteers played a race game and a flight game in a head-mounted display. About once per second the game logged speed, acceleration, three camera-rotation components (range $[-1, 1]$; the parameterization is not documented), frame rate, region of interest, configuration flags and the self-reported discomfort level (0–3: none, slight, moderate, severe) [20], [34]. Each session also stores a pre-play Cybersickness Profile Questionnaire (CSPQ: gender, age band, VR experience, glasses, vision problems, eye dominance, flicker sensitivity). We rebuilt every input from the raw XML exports: the dataset authors' curated release (35 seated sessions, December 2019; 9,390 samples) matches 35 raw exports sample by sample and is the evaluation set; the 37 October-2019 standing sessions are used only to debug code and calibrate training budgets (validation cost plateaued after 20–30 epochs, so 40 epochs were fixed).

The target is $y_t = 1$ (event) if the discomfort level at second t is at least 1. The prevalence is 44.2 % (Flight 41.2 %, Race 48.1 %); grades 2–3 occur in 1,450 of 3,993 Race but only 41 of 5,397 Flight samples. Labels are near step functions: the binary label changes 1.1 times per session on average, never decreases in 23 of 35 sessions, and 9 sessions never report discomfort. The prompting procedure is not documented, so consecutive labels are a held self-report, not independent measurements. No session was attacked: y marks discomfort, not a manipulation.

B. SESSION-LINKAGE GROUPS (STAGE 3)

Consecutive seconds of a session are near-duplicates with the same label, so every split must keep each session—and, as far as the data allow, each participant—on one side. The release has no participant identifier, and some participants restarted after an aborted session. We link two sessions if (a) their nine profile fields are identical or (b) one starts within 180 s after an aborted session (< 200 samples) and their profiles differ in at most two fields. The connected components are 21 session-linkage groups, the unit of every split. The rule may merge different people with the same profile, which is conservative, but cannot link one person who answered the questionnaire differently in non-consecutive sessions (Section V-D).

C. CAUSAL FEATURES AND TRAIN-ONLY PREPROCESSING (STAGE 5)

A feature is causal if its value at second t uses only seconds $0, \ldots, t$ of the same session; this was verified by recomputing all features on sessions truncated at random times. The 39 telemetry features are 13 instantaneous values (game, elapsed time, speed, acceleration, |acceleration|, the three rotation components q_t, $\|q_t - q_{t-1}\|$, frame rate, three region-of-interest indicators), 22 trailing-window statistics over 5 s and 15 s (mean, SD and change of speed; mean and maximum |acceleration|; spike rate; mean and SD of rotation change; mean, minimum and drop fraction of frame rate) and 4 cumulative values (rotation change, spikes, frame drops, running mean speed). A spike is |acceleration| above the 90th percentile of the training rows; a frame drop is a frame rate below 80 fps. Eleven questionnaire indicators ("profile") encode the seven CSPQ items. The spike threshold, the imputation of the logger's zero frame rate on some first samples, and a median/IQR scaler are fitted on the training rows of each inner split and applied unchanged to its validation and test rows. Constant columns (field of view and two camera flags per game, four configuration flags, posture) and the absolute position (a proxy for elapsed course distance) are removed; no oversampling or synthetic data are used. The feature set is a hyper-parameter: telemetry (39), telemetry+profile (50), exposure (8 a-priori exposure descriptors: game, elapsed time, cumulative rotation change and spikes, running and 15-s mean speed, 15-s means of rotation change and |acceleration|) or exposure+profile (19).

D. DECISION FORMULATION: MDP OR CONTEXTUAL BANDIT?

Each session is an episode. At second t the detector observes the causal feature vector $s_t \in \mathbb{R}^d$, chooses $a_t \in \{0, 1\}$ (silent, alarm) and receives $r_t = R(a_t, y_t)$ with

$$\begin{aligned} R(1,1) &= +0.5, \quad R(0,0) = +0.1, \\ R(0,1) &= -0.5, \quad R(1,0) = -0.1. \end{aligned} \tag{1}$$

The next state s_{t+1} is the next second of the same session; the last second is terminal ($d_t = 1$). Because the telemetry

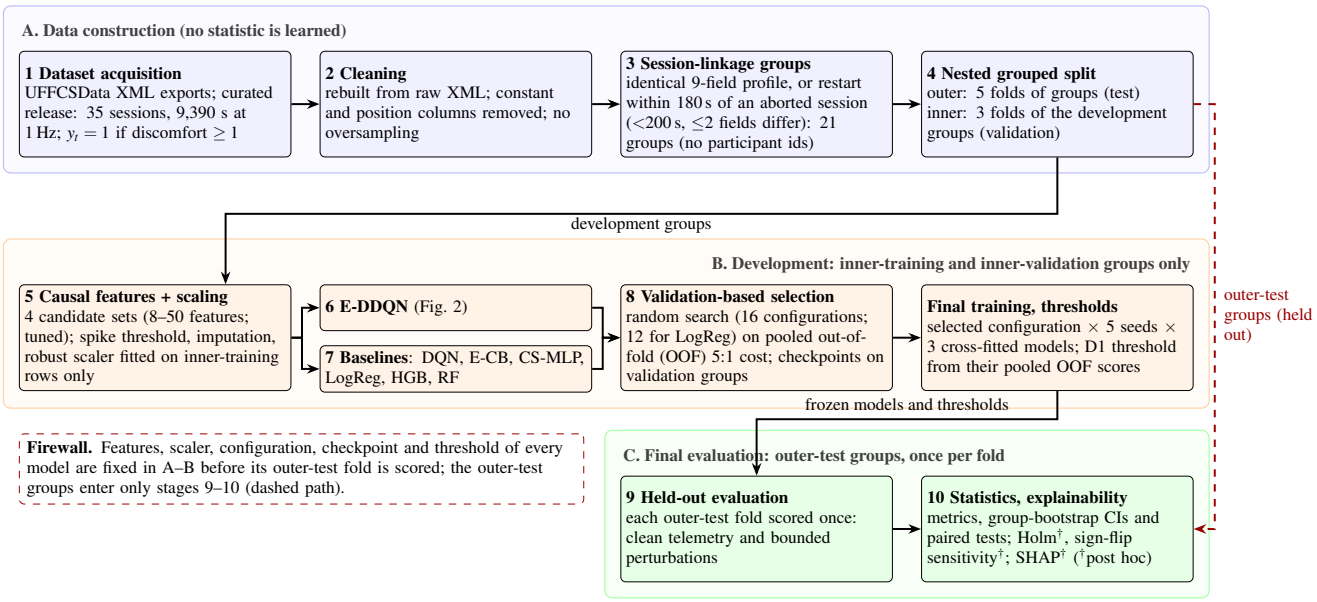


FIGURE 1. Implemented workflow. A: data construction; B: development on inner folds only (E-DDQN in Fig. 2); C: single evaluation of each outer-test fold (protocol in Fig. 3). The dashed path marks the outer-test groups, which bypass every fitted step.

was logged without an alarm in the loop, $P(s_{t+1} \mid s_t, a_t) = P(s_{t+1} \mid s_t)$, and for any $\gamma < 1$

$$Q^*(s,a) = \mathbb{E}[R(a,y_t) \mid s_t{=}s] + \underbrace{\gamma\,\mathbb{E}[(1-d_t)V^*(s_{t+1}) \mid s_t{=}s]}_{\text{same for both actions}}. \quad (2)$$

Hence (i) the logged-data problem is a contextual bandit [35]: the best action maximizes the immediate expected reward for every γ. (ii) With $p(s) = P(y = 1 \mid s)$ the optimal decision margin is

$$Q^*(s,1) - Q^*(s,0) = p(s)\,\Delta_{\mathrm{FN}} - (1-p(s))\,\Delta_{\mathrm{FP}}, \quad (3)$$

with regrets

$$\Delta_{\mathrm{FN}} = R(1,1) - R(0,1) = 1.0, \quad \Delta_{\mathrm{FP}} = R(0,0) - R(1,0) = 0.2, \quad (4)$$

so the greedy policy alarms iff $p(s) \geq \Delta_{\mathrm{FP}}/(\Delta_{\mathrm{FN}} + \Delta_{\mathrm{FP}}) = 1/6$: exactly the Bayes rule for costs $C_{\mathrm{FN}} : C_{\mathrm{FP}} = 5 : 1$ [17]. The asymmetric reward encodes an operating point, not a different objective. (iii) Solving Eq. (3) for p maps a Q-function to a score on the nominal scale of $P(y = 1 \mid s)$,

$$\hat{p}(s) = \frac{\Delta Q(s) + \Delta_{\mathrm{FP}}}{\Delta_{\mathrm{FN}} + \Delta_{\mathrm{FP}}}, \qquad \Delta Q(s) = Q(s,1) - Q(s,0), \quad (5)$$

computed with each agent's own reward table ($\Delta_{\mathrm{FN}} = \Delta_{\mathrm{FP}} = 2$ for the ± 1 reward); $\hat{p}$ is an uncalibrated estimate and can leave $[0,1]$. These consequences follow from known results [17], [19]; we state them because asymmetric-reward DQN detectors are usually presented without this link.

Score scale. Class-balanced replay or minibatches (a share $1-\rho$ or one half of events instead of the training prevalence π, 0.405–0.483) multiply the odds estimated by the scores by $k = \frac{1-\rho}{\rho}\frac{1-\pi}{\pi}$, so the nominal cut-off $\hat{p} \geq 1/6$ corresponds to $P(y = 1 \mid s) \geq 1/(1+5k)$: 0.084–0.157 for E-DDQN (approximate, since its γ of 0.9–0.99 lets some residuals exceed the Huber threshold), 0.120–0.208 for E-CB and 0.121–0.149 for the balanced CS-MLP. The Huber loss with the standard DQN's ± 1 reward makes its $\hat{p}$ a nonlinear function of $P(y = 1 \mid s)$ (cut-off 0.25 at $\gamma = 0$). These shifts affect the untuned rule D0; the primary rule D1 and the ranking metrics choose thresholds on the scores themselves, and rescaling every inner model to $P(y = 1 \mid s)$ changes at most 0.8 % of D1 decisions, the cost by at most 0.007 and ROC-AUC by at most 0.003 (Supplementary Sec. S5.12).

Episode termination and discounting. Earlier DQN classifiers end an episode when a minority sample is misclassified [11], [12]; a miss then also forfeits $\gamma V(s_{t+1})$, an extra γ-dependent penalty, so the problem is no longer a bandit. E-DDQN ends episodes only with the session, so Eqs. (2)–(5) apply. Bootstrapping ($\gamma > 0$) only adds the action-independent term of Eq. (2) and its estimation noise; we therefore tune $\gamma \in \{0, 0.5, 0.9, 0.99\}$ on validation data and also evaluate the separately tuned $\gamma = 0$ special case (E-CB). Adaptive sequential control would require an intervention in the loop, which the data do not contain, so we make no such claim.

E. E-DDQN (STAGE 6)

Fig. 2 shows E-DDQN as implemented; Table 1 lists its settings and search space. A multilayer trunk (Linear, batch or layer normalization [36], ReLU and dropout [37] except after the last layer) feeds a dueling head [14], $Q(s,a) = V(s) + A(s,a) - \frac{1}{2}\sum_{a'} A(s,a')$. The Double-

Algorithm 1 E-DDQN training for one inner split

Initialize θ, $\theta^- \leftarrow \theta$, buffer $\mathcal{D}$; optional warm-up with random actions
for epoch $= 1, \ldots, 40$ **do**
  Concatenate training sessions in random order, each in temporal order; last second of each session terminal
  for each chunk of 256 consecutive transitions **do**
    $a_t \leftarrow \varepsilon$-greedy on $Q_\theta(s_t, \cdot)$; $r_t \leftarrow R(a_t, y_t)$
    store $(s_t, a_t, r_t, s_{t+1}, d_t, y_t)$ [and $(s_t, 1-a_t, R(1-a_t, y_t), \ldots)$ if full-information]
    every 2 chunks: 1, 2 or 4 updates on class-balanced minibatches with Eq. (6), Huber loss, Polyak update
  end for
  5 consolidation updates; decay ε; cosine LR step
  every 2 epochs: keep θ if the validation 5:1 cost improves
end for
return checkpoint; score $\hat{p}(s)$ by Eq. (5)

DQN target [13] is

$$y_t = r_t + \gamma(1-d_t)\, Q_{\theta^-}\big(s_{t+1}, \arg\max_{a'} Q_\theta(s_{t+1}, a')\big), \quad (6)$$

with the arg-max taken by the online network in evaluation mode ($y_t = r_t$ for $\gamma = 0$). The Huber loss [38], [39] between $Q_\theta(s_t, a_t)$ and y_t is minimized with Adam [40], weight decay, gradient-norm clipping and cosine learning-rate decay [41]. After each update $\theta^- \leftarrow (1-\tau)\theta^- + \tau\theta$ [42], and batch-normalization statistics are copied (all five selected E-DDQN networks used layer normalization, so this copy was inactive in the reported runs). Minibatches from the replay buffer [43] (50,000 transitions; 100,000 with full-information replay) contain a fixed share ρ of non-event and $1-\rho$ of event transitions. Actions are ε-greedy, with an optional warm-up of random actions. With full-information replay (tuned on/off) the action not taken is also stored with its own reward—a valid transition that the labels allow offline. Every second epoch the inner-validation groups are scored and the epoch with the lowest achievable 5:1 cost is kept (Algorithm 1).

Implementation checks. An earlier implementation of this study built transitions from shuffled rows without terminal flags, stored the label as the action while rewarding a different action, and averaged only parameters into the target network. Unit tests guard against all three. On the real folds, every one of the 240,646 transitions written to the replay buffer in shortened runs of E-DDQN, E-CB and the standard DQN with each fold's selected configuration was logged: each stored reward is that of the stored action and the label of its state, each non-terminal next state is the next second of the same session, and terminal flags occur exactly at session ends (Supplementary Sec. S8).

F. BASELINES (STAGE 7)

Every baseline receives the same splits, candidate feature sets, search procedure and number of configurations (16; 12 for logistic regression), checkpoint rule (neural models), cross-fitting and threshold rules (Table 1). The number of configurations is matched, not the coverage: the search spaces range from about 3.0 million configurations (E-DDQN) to 192 (random forest; Supplementary Sec. S2.1). The **standard DQN** [32] uses a symmetric ± 1 reward, a plain head, the max-operator target, periodic hard target copies and uniform replay; **E-CB** is E-DDQN with $\gamma = 0$; the **CS-MLP** is a multilayer perceptron with event-weighted cross-entropy ($w = 5$); **logistic regression**, **histogram gradient boosting (HGB)** and a **random forest (RF)** [44] use class weight 5. A weighted classifier estimates $p_w = 5p/(5p+1-p)$, which is mapped back by

$$\hat{p} = \frac{p_w}{5 - 4p_w}, \quad (7)$$

so that all models share the nominal scale of Eq. (5). Always alarm and never alarm are trivial references. Because exposure duration drives cybersickness [45] and most selected configurations used exposure features, two exposure-time baselines were specified after the primary analysis and run under the same protocol: **Time-LR** (logistic regression on elapsed time) and **Time+game HGB**.

TABLE 1. Implementation and training settings (`configs/protocol.json`, except the rows marked †; braces: searched values).

Item	Setting
Splits	outer 5 / inner 3 stratified group folds (game × label); seeds 20261012 / 20261013–17
Search	random, 16 configurations per family (LogReg 12 of 24), drawn once (sampling seed 777 + family index; training seed 777); objective: pooled OOF 5:1 cost at its best threshold, ties PR-AUC
Neural (all)	feature set {4 sets}; trunk {64–64, 128–128, 256–256, 256–512–256}; dropout {0, .1, .2, .3}; norm {batch, layer}; learning rate η {3e-4, 1e-3, 3e-3}, cosine to 1e-5; weight decay {0, 1e-4, 1e-3}; batch {256, 512}; 40 epochs; checkpoint every 2 epochs
Agents (all)	Huber loss (threshold 1), Adam, grad.-norm clip 1.0; buffer 50,000 (100,000 with full information); chunks of 256, learning step every 2 chunks with {1, 2, 4} updates, 5 consolidation updates per epoch; ε 1.0→0.01, decay {0.90, 0.95, 0.97}/epoch; warm-up {0, 4,000}
E-DDQN	reward Eq. (1); dueling; double; soft target τ {0.005, 0.01, 0.05}; γ {0, 0.5, 0.9, 0.99}; balanced replay ρ {0.4, 0.5, 0.6}; full-information {off, on}
E-CB	as E-DDQN with $\gamma = 0$; ρ, full-information searched
DQN	± 1 reward; plain head; max target; hard copy every {10, 100, 500} updates; γ {0, 0.5, 0.9, 0.99}; uniform replay; no full-information replay
CS-MLP	weighted cross-entropy ($w = 5$); minibatches {uniform, balanced}
LogReg	C {1e-4, …, 10} (6 values)
HGB	η {.02, .05, .1}; iterations {50, 100, 200, 400}; leaves {4, 8, 16, 31}; min. leaf {20, 100, 300}; ℓ_2 {0, 1, 10}
RF	300 trees; depth {3, 5, 8, none}; min. leaf {5, 25, 100, 300}; features {sqrt, 0.5, 1.0}
Final	5 seeds (1000–1004) × 3 cross-fitted inner models per outer fold
Inference†	group bootstrap, 2,000 replicates (analysis code committed with the protocol); exact sign-flip test over 2^{21} sign patterns (post hoc)
Software†	Python 3.13, PyTorch [46] 2.14.1 (run on the CPU), scikit-learn 1.9.1; one thread per process; full pipeline about 2.5–3.5 h on four CPU cores; [CPU model and memory to be added]

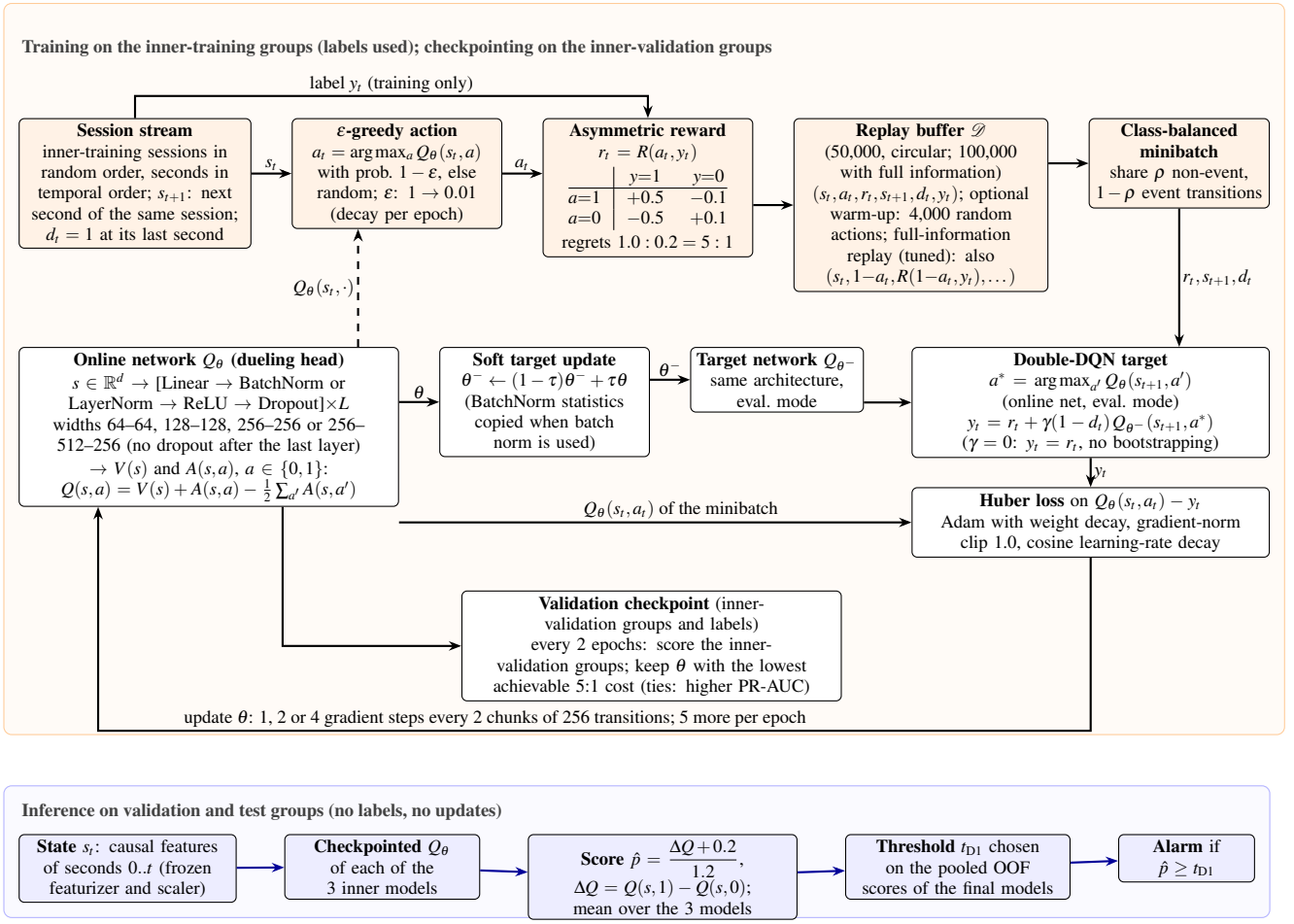


FIGURE 2. E-DDQN as implemented. Top (orange): training on the inner-training groups, where labels define the reward; bottom (blue): inference on validation and test groups with the checkpointed networks, the score of Eq. (5) and the validation-chosen threshold of rule D1. Dashed: the online network selects actions; $\gamma = 0$ removes the target-network path.

G. EVALUATION PROTOCOL AND STATISTICS (STAGES 4 AND 8–10)

Fig. 3 shows the protocol; its split, search, checkpoint, threshold and seed rules were committed before any model was trained on the evaluation sessions (the post-hoc additions are marked). The 21 groups are split into 5 outer folds; each serves once as test data and the rest form its development set, which is split into 3 inner folds so that every development second receives one out-of-fold (OOF) validation prediction. For each family and outer fold: (1) each configuration is trained on the three inner splits with one seed, and the one with the lowest pooled-OOF 5:1 cost $c = (5\,\mathrm{FN} + \mathrm{FP})/N$ is selected; (2) it is retrained with five seeds, keeping the three checkpointed inner models per seed (cross-fitting), whose mean is the test score; (3) thresholds are chosen on the pooled OOF scores of these final models: the primary rule D1 minimizes the 5:1 cost (ties: closest to $1/6$); secondary rules are D0, the fixed cut-off $\hat{p} \geq 1/6$ (the greedy policy for the asymmetric agents), and an OOF recall of at least 0.95. Each OOF score comes from one inner model, whereas each test score averages three, so averaging can make test scores less spread than the scores on which the threshold was chosen (checked post hoc; Section V-D, limitation (ix)); (4) the outer-test fold is scored once, on clean and on perturbed telemetry. Checks on the real folds confirm that no session, linkage group, exact questionnaire profile, raw telemetry row or feature vector crosses a partition boundary in any of the 15 outer $\times$ inner splits, that preprocessing is unaffected by evaluation rows, and that every configuration, checkpoint and threshold is reproduced from inner-validation data alone (Supplementary Sec. S8). This establishes session- and linkage-group-disjoint evaluation, not participant independence.

Seed means and seed ensemble. We report mean $\pm$ SD over the five seeds. Analyses that need one prediction per sample (paired tests, confidence intervals, comparison with always alarming, event-level metrics, matched operating points, perturbations, cost curves) use the seed ensemble, the mean score over seeds thresholded on the mean OOF score; this was fixed in the evaluation code before the evaluation. The two can differ: E-DDQN's ensemble cost is 0.726 against a seed mean of 0.663, the standard DQN's

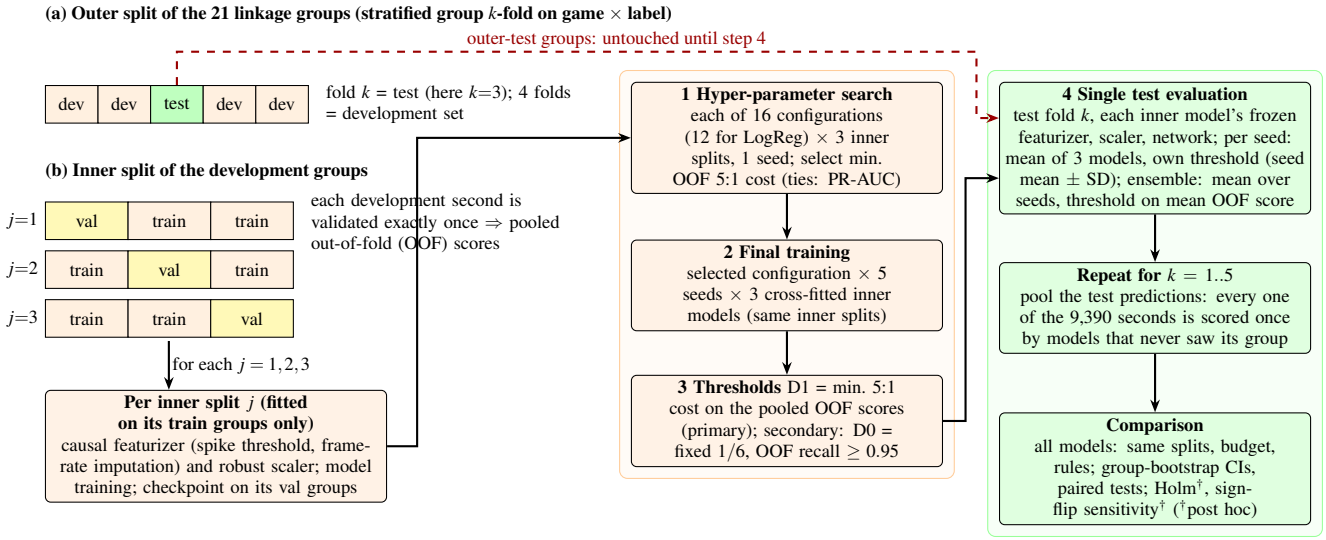


FIGURE 3. Nested, session-linkage-grouped evaluation protocol, applied identically to every model. Within each outer fold, all fitting, selection, checkpointing and thresholding use inner folds of the development groups; the outer-test groups are used once, in step 4.

0.536 against 0.584. No seed was selected or discarded.

Metrics and inference. We report recall, specificity, balanced accuracy, MCC [47], missed events, false positives, the 5:1 cost, ROC-AUC and PR-AUC [48] (further metrics in Supplementary Sec. S3.1) and event-level alarm fractions. Confidence intervals are percentile intervals of a bootstrap over the 21 linkage groups (2,000 replicates); they describe the sampling of test groups given the trained models, not retraining variability beyond the five seeds. The primary paired test is a group bootstrap of the seed-ensemble cost difference, $p = 2\min\{\Pr(\Delta^* \le 0), \Pr(\Delta^* \ge 0)\}$ (resolved to about 0.001); it was implemented in the analysis code committed with the frozen protocol and yields effect size, CI and p-value from one distribution. An exact group sign-flip test over all 2^{21} sign patterns, added after a pre-submission review, is the sensitivity analysis. Every comparison is reported as an effect size with its 95 % CI and Holm-adjusted [49] p-values, and a difference is treated as firmly established only if it is significant under both procedures. With 21 unequal groups, cluster-bootstrap inference is approximate and can be anti-conservative [50], so p-values between 0.01 and 0.05 are read as approximate. Sample-level McNemar tests [51], [52] are secondary because consecutive seconds are dependent.

Post-hoc analyses (Supplementary Sec. S5). The following were specified after the primary results were known and committed before being run, unless stated otherwise: the exposure-time baselines; Holm adjustment of the six comparisons with E-DDQN; comparison with always alarming (its Holm adjustment added without prior commitment); session-level bootstrap and leave-one-group-out jackknife; validation-chosen cost curves over cost ratios R from 1 to 20; within-fold ROC-AUC, sign-flip tests and feature-set tallies (after review, with all results known); a search-draw experiment, specified with three decision rules and committed before it was run, which re-tunes E-DDQN and the standard DQN with two further random draws of 16 configurations and one exposure-only draw (Supplementary Sec. S5.11); the descriptive score-scale calculation above (added in the final pre-submission check); and a threshold-scale check of the D1 results (specified and committed before it was run; Supplementary Sec. S9.10). The paired comparisons of the exposure-time baselines with the primary models and the Q-value-scale diagnostic were added without prior commitment.

H. TELEMETRY-PERTURBATION STRESS TESTS AND EXPLAINABILITY

The motivating threats degrade frame pacing, distort camera or locomotion motion, or manipulate the field of view [2]–[4]. The data contain no attacked sessions, so we use bounded, synthetic changes of the outer-test telemetry as a stress test of which inputs the detectors use; features are recomputed with the frozen featurizer and scaler, and labels are unchanged. The 24 variants, fixed before evaluation, are frame-rate capping, orientation jitter, speed up-scaling and acceleration-spike injection at three strengths, applied to the whole session or one 60-s window (values and bounds in Supplementary Sec. S4.1; the 60- and 45-fps caps lie below every Flight frame rate after the first second, 70–94 fps). Field-of-view manipulation is not evaluated because it never varies within a game. An adaptive adversary with label knowledge picks, per session, among 17 whole-session variants (including five "hiding" manipulations that mask stress signals: speed down-scaling, orientation smoothing, frame-rate restoration) the one that minimizes alarms on event seconds or maximizes alarms on non-event seconds; these are lower bounds, weaker than gradient-based attacks [4],

[31]. Post hoc, SHAP values [53] of the decision margin ΔQ are computed for one separately retrained E-DDQN model per outer fold with the expected-gradients estimator (DeepExplainer [54] failed its additivity check on the layer-normalized networks; Supplementary Sec. S9.8).

IV. RESULTS

A. EFFECT OF THE SPLIT PROTOCOL

Table 2 evaluates the same 21 raw columns under three protocols with an untuned gradient-boosting classifier, quantifying a known pitfall [15], [26] on these data. Oversampling every discomfort grade to 5,243 samples before a 70/15/15 row split, the protocol of the earlier version of this study, gives 97.6 % accuracy and ROC-AUC 0.995; 92 % of the positive test rows have an exact duplicate in training. A row-level split without oversampling still gives ROC-AUC 0.992, whereas a session-grouped split gives 0.523 (MCC 0.005). The earlier E-DDQN implementation behaves alike (MCC 0.870 vs. −0.004). Near-perfect scores on these data thus measure recognition of seen sessions, without any RL; all further results use the grouped protocol of Fig. 3.

B. SESSION-GROUPED COMPARISON UNDER THE 5:1 COST (RQ2)

Table 3 gives every model's result, with thresholds chosen on validation groups only. Two facts dominate. First, discrimination is weak once sessions and groups are not shared: the best primary ROC-AUC is 0.692 (HGB) and 0.688 (RF), and the best PR-AUC 0.595 (DQN) against a prevalence of 0.44. Second, under a 5:1 cost every cost-optimal operating point lies near "always alarm" (cost 0.558): recall is high, but 82–97 % of non-event seconds trigger an alarm. The random forest has the lowest cost (seed mean 0.491±0.003; ensemble 0.485; point estimates, differences among supervised models not tested).

With its primary search draw (draw 1), E-DDQN has the highest cost and lowest ROC-AUC of the 7 primary models (point estimates; tests in Section IV-C): cost 0.663±0.048 (ensemble 0.726), recall 93.4 %, specificity 7.4 %, ROC-AUC 0.538 (ensemble 0.541, 95 % CI 0.382 to 0.690, which includes chance). Its seed-ensemble cost exceeds the random forest's in 5 of 5 outer folds, by 0.004 to 0.727. The standard DQN reaches cost 0.584 and ROC-AUC 0.661, E-CB 0.575 and 0.653.

Exposure-time baselines (post hoc). Time-LR, a logistic regression on elapsed time alone, reaches ROC-AUC 0.704 (Fig. 4). Computed within each outer fold and averaged (within-fold ROC-AUC, which avoids pooling the scores of five fold-specific models), the random forest is slightly higher (0.715 vs. 0.709) and E-DDQN lowest (0.539). Time-LR's ROC-AUC does not differ significantly from the random forest's (difference +0.016, CI −0.071 to +0.102), and the wide interval does not establish equivalence either. At the 5:1 operating point the random forest has a lower cost than Time-LR (0.485 vs. 0.548; group bootstrap $p = 0.021$, sign-flip $p = 0.044$; approximate), so the evidence that its additional inputs help is suggestive, not conclusive. Time+game HGB (cost 0.517, ROC-AUC 0.676) does not differ significantly from the random forest either (cost $p = 0.274$, ROC-AUC $p = 0.760$). Time-LR ranks seconds better than the primary E-DDQN (ROC-AUC difference +0.163, CI +0.063 to +0.287). No primary model was shown to discriminate better than a population onset curve—a risk of discomfort that rises with time in the session—as long documented for exposure duration [45]; modest gains remain possible within the wide intervals.

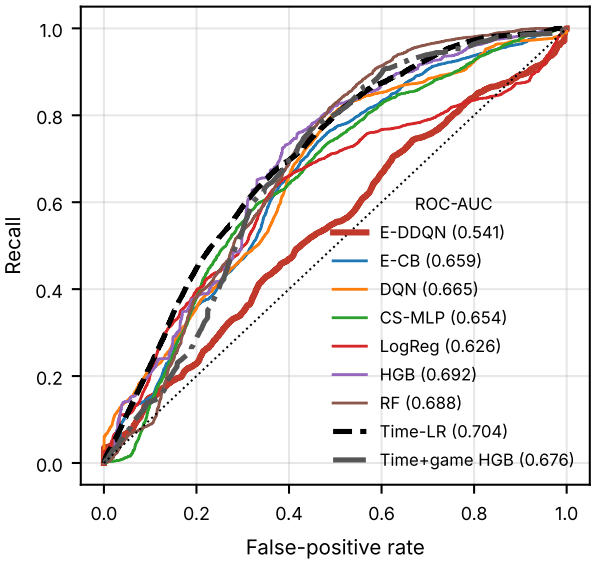


FIGURE 4. ROC curves of the seed-ensemble scores, pooled over the outer-test folds, for the primary models and the exposure-time baselines (dashed). Legend: ensemble ROC-AUC.

C. STATISTICAL COMPARISON

Table 4 compares E-DDQN with each primary model on identical test samples (seed ensembles). E-DDQN's ensemble cost exceeds the others' by +0.175 to +0.242; these differences are larger than those of seed means (+0.079 to +0.172) because E-DDQN's ensemble is worse than its average seed. After Holm adjustment, only the difference from the random forest is significant under the primary group bootstrap (+0.242, 95 % CI +0.052 to +0.528; $p = 0.006$); the other five adjusted p-values lie between 0.070 and 0.105, and the difference from the standard DQN (+0.190) is not significant (unadjusted $p = 0.079$). The sign-flip sensitivity analysis gives $p = 0.083$ (Holm) for the random-forest comparison. The direction is stable—the session-level bootstrap gives the same pattern (Holm $p = 0.006$) and no leave-one-group-out deletion changes the sign (+0.137 to +0.284)—but because the difference is significant under the primary procedure only, it is not firmly established, and we do not conclude that the random forest is superior.

Against always alarming, no model's cost difference is significant after Holm adjustment with linkage groups as

TABLE 2. Effect of the evaluation protocol on the same data (21 raw columns; test-split metrics). "Proxy": reconstruction of the earlier protocol, whose 41-feature input files were not available.

Model	Evaluation protocol	Acc.	Recall	Spec.	MCC	ROC-AUC
HGB	Oversample to 5,243/grade, then 70/15/15 row split (earlier protocol)	97.6	99.8	91.0	0.935	0.995
HGB	Row-level random split, no oversampling	95.3	94.1	96.3	0.905	0.992
HGB	Grouped 5-fold split (session-linkage groups)	52.2	33.6	66.9	0.005	0.523
E-DDQN (earlier code)	Earlier script, earlier protocol (proxy)	95.2	99.2	83.1	0.870	–
E-DDQN (earlier code)	Earlier script, grouped split (proxy)	41.1	92.1	7.7	−0.004	–

TABLE 3. Session-grouped comparison (rule D1; 9,390 outer-test samples; recall, specificity, balanced accuracy in %). Left: mean ± SD over five seeds. Right: seed ensemble, used for all paired tests and confidence intervals; FN and FP are exact counts. Bold: proposed model and lowest primary cost. HGB, logistic regression and the post-hoc baselines are deterministic (SD = 0). Further metrics: Supplementary Sec. S3.1.

	Mean ± SD over five seeds							Seed ensemble (paired tests)				
Model	Recall	Spec.	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	Recall	Spec.	$\text{Cost}_{5:1}$	FN	FP
E-DDQN	93.4±2.5	7.4±2.7	50.4±1.2	0.016±0.054	0.663±0.048	0.538±0.016	0.483±0.019	90.4	8.1	0.726	400	4,820
E-CB	97.1±2.9	8.4±2.8	52.8±1.2	0.122±0.052	0.575±0.055	0.653±0.011	0.562±0.006	96.6	14.6	0.552	140	4,480
DQN	95.2±2.7	14.5±4.9	54.8±1.6	0.158±0.038	0.584±0.040	0.661±0.013	0.595±0.022	96.9	16.3	0.536	129	4,391
CS-MLP	99.4±0.7	2.8±1.9	51.1±0.8	0.072±0.049	0.557±0.012	0.644±0.018	0.521±0.022	99.4	6.1	0.537	23	4,925
LogReg	99.1±0.0	8.1±0.0	53.6±0.0	0.167±0.000	0.532±0.000	0.626±0.000	0.576±0.000	99.1	8.1	0.532	36	4,817
HGB	97.0±0.0	14.6±0.0	55.8±0.0	0.197±0.000	0.543±0.000	0.692±0.000	0.593±0.000	97.0	14.6	0.543	124	4,479
RF	98.5±0.3	18.2±0.7	58.3±0.3	0.266±0.004	**0.491±0.003**	0.688±0.002	0.548±0.001	98.7	18.3	**0.485**	54	4,281
Post-hoc exposure-time baselines (added after the primary analysis)												
Time-LR (elapsed time)	96.7±0.0	15.0±0.0	55.8±0.0	0.194±0.000	0.548±0.000	0.704±0.000	0.605±0.000	96.7	15.0	0.548	138	4,457
Time+game HGB	97.4±0.0	17.6±0.0	57.5±0.0	0.239±0.000	0.517±0.000	0.676±0.000	0.550±0.000	97.4	17.6	0.517	107	4,319
Always alarm	100.0	0.0	50.0	0.000	0.558	–	–	100.0	0.0	0.558	0	5,243
Never alarm	0.0	100.0	50.0	0.000	2.208	–	–	0.0	100.0	2.208	4,147	0

units (random forest −0.074, CI −0.135 to −0.019, adjusted $p = 0.108$; E-DDQN +0.168, adjusted $p = 0.728$). This reflects the strength of the evidence, not demonstrated equivalence; resampling sessions instead, which ignores dependence between linked sessions, leaves the random forest's advantage significant (adjusted $p = 0.018$, approximate).

D. OPERATING POINTS AND COST RATIOS

With the untuned rule D0 (for E-DDQN the greedy policy, an effective cut-off of 0.084–0.157 on $P(y = 1 \mid s)$) E-DDQN reaches recall 77.2 % at specificity 28.0 % (cost 0.905), the random forest 94.5 % at 34.9 % (0.486); under a validation recall of at least 0.95 the random forest again has the lowest cost (0.483; E-DDQN 0.785; Supplementary Sec. S9.3). At validation-matched recall targets, E-DDQN's thresholds transfer worst (Supplementary Sec. S9.4). Since the reward encodes the same operating point as a cost-weighted classifier (Eq. (3)), on these data it provided no advantage that cost weighting plus a validation-chosen threshold did not.

The cost curves (Fig. 5) show that this negative result does not depend on the 5:1 ratio. At $R = 1.5$–2, the relative costs of all models except E-DDQN lie between 0.72 and 0.91; as R grows, always alarming becomes nearly optimal and models that still miss events rise above 1 (at $R = 20$: E-DDQN 1.508, DQN 1.304). The lowest relative cost over all models and ratios is 0.72 (random forest, $R = 1.5$), so no model removes more than about 28 % of the cost of the better trivial policy at any ratio (point estimates).

Event level. At the cost-optimal operating points every model flags almost every session: E-DDQN alarms on

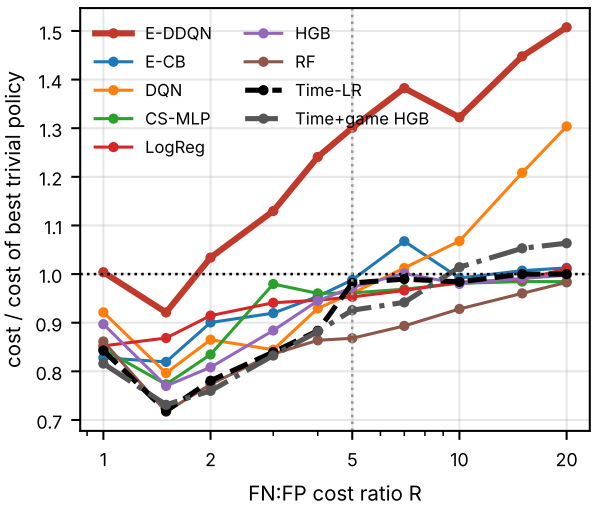


FIGURE 5. Validation-chosen cost curves (post hoc): for each cost ratio R, the threshold minimizing $R\,\text{FN} + \text{FP}$ on validation predictions is applied to the test fold; the seed-ensemble test cost is divided by that of the better trivial policy. Values below 1 beat both trivial policies; dotted line: $R = 5$.

96.6 % of the seconds of the 9 event-free sessions and 82.3 % of pre-onset seconds (random forest 75.3 % and 61.3 %), so events are "detected" with zero median delay by construction, not by discrimination.

TABLE 4. Paired comparisons with E-DDQN (A) on the same 9,390 test samples (seed ensembles, rule D1). b: A correct, B wrong; c: A wrong, B correct. ΔCost: difference of seed-ensemble costs; CI and p_{group} from 2,000 group-bootstrap replicates (unadjusted; Holm-adjusted values in the text); p_{exact}: sample-level McNemar test, which ignores within-session dependence.

B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\text{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	299	899	5,220	4,620	299.5	<0.001	+0.175 [+0.016, +0.448]	0.014
DQN	297	997	5,220	4,520	377.6	<0.001	+0.190 [−0.017, +0.493]	0.079
CS-MLP	402	674	5,220	4,948	68.3	<0.001	+0.190 [+0.005, +0.485]	0.039
LogReg	389	756	5,220	4,853	117.0	<0.001	+0.194 [+0.011, +0.477]	0.026
HGB	203	820	5,220	4,603	370.9	<0.001	+0.183 [+0.011, +0.460]	0.035
RF	293	1,178	5,220	4,335	531.2	<0.001	+0.242 [+0.052, +0.528]	0.001

E. WHY E-DDQN UNDER-PERFORMS IN THE PRIMARY ANALYSIS

Validation selected $\gamma = 0.99, 0.99, 0.99, 0.99, 0.9$ for E-DDQN's five folds and the questionnaire indicators in 5 of 5 folds.

Search-draw experiment (specified and committed before it was run). E-DDQN and the standard DQN were re-tuned under the full protocol with two further random draws of 16 configurations and one exposure-only draw, with three decision rules fixed in advance (R1: is the E-DDQN–DQN cost comparison overturned? R2: does E-DDQN's result depend on the draw? R3: does restricting both to exposure features close their ranking gap?). In none of the three full-space draws or the exposure-only draw did E-DDQN have a significantly lower cost than the DQN (seed-ensemble differences −0.135 to +0.190; smallest $p = 0.079$, in draw 1; R1). E-DDQN's own result depended strongly on the draw (R2): seed-mean cost 0.549–0.663 and within-fold ROC-AUC 0.539, the lowest of the primary models, to 0.714, close to the highest (RF 0.715; not tested); the DQN varied less (0.629–0.666). The ranking of the two agents reversed: the DQN ranked seconds better with draw 1 (unadjusted $p = 0.021$, approximate), E-DDQN with draw 2 (unadjusted $p = 0.009$). Restricted to exposure features, E-DDQN's within-fold ROC-AUC (0.687) was within 0.026 of the DQN's (R3).

Search draw and selection (exploratory, post hoc). Each family's configurations were drawn once and reused in every outer fold; E-DDQN's primary draw offered only 2 exposure-only configurations (DQN 6; Supplementary Sec. S5.10), and validation selected configurations with questionnaire indicators in 5 of 5 folds. The selection criterion is nearly flat near always alarming (0.42–0.54 for the selected configurations vs. 0.558), so with noisy training curves and validation checkpointing configurations can look best on validation folds by chance [55]. The selection optimism, outer-test cost minus the inner-validation cost at selection, is +0.288 for E-DDQN and −0.015–+0.070 for the other models (Supplementary Sec. S9.5).

Questionnaire indicators (exploratory). Across the seven primary models, the 13 selected configurations with questionnaire indicators rank worse within folds than the 22 without them (deviation from the fold mean −0.069 vs. +0.041; without E-DDQN −0.056 vs. +0.020; Supplementary Sec. S5.9). With 21 groups such indicators may fit group-level label rates that do not transfer; feature sets were selected, not assigned, so this suggests but does not establish a cause.

Bootstrapping. With E-DDQN's configurations fixed, $\gamma = 0$ raises ROC-AUC only from 0.538 to 0.576 (Table 5), and in draw 2 E-DDQN reached its best ranking with $\gamma = 0.99$ in 3 of 5 folds. Q-values stay finite but exceed the return bound $r_{\max}/(1-\gamma)$ by up to a factor of 2.3 (Supplementary Sec. S5.6).

Together these results suggest that sensitivity to the search draw, combined with noisy validation-based selection, contributes substantially to E-DDQN's weak primary result. They do not isolate a single cause—draw, feature sets, discount factors and other hyper-parameters varied together—and they do not show that any E-DDQN component helps a configuration that ranks well.

F. COMPONENT ABLATION AND REWARD RATIO

Table 5 removes or changes one component of E-DDQN at a time while keeping the configuration selected for each outer fold fixed (five seeds; same protocol). No formal test was performed, so the differences below are point estimates. Because the selected configurations already used no warm-up in four of the five folds, the "no warm-up" variant differs from the full model only in the remaining fold. This analysis and the reward-ratio analysis below hold E-DDQN's primary (draw-1) configurations fixed, which rank weakly (Section IV-E); they cannot show whether a component helps a configuration that ranks well. Every variant ranks seconds only weakly (ROC-AUC 0.529–0.576, where 0.5 is chance). No single change moves the cost by more than about one seed standard deviation of the full model (0.663±0.048). The lowest cost is obtained by replacing the asymmetric reward with the symmetric ±1 reward (0.625) and the highest ROC-AUC by switching off bootstrapping ($\gamma = 0$; 0.576); hard target copies (0.634), no dueling head (0.644), uniform replay (0.660), no warm-up (0.656) and toggling full-information replay (0.667) leave the cost essentially unchanged, and removing the double estimator slightly increases it (0.681). None of these components can be credited with a gain in safety or accuracy. Every variant is scored on the same nominal implied-probability scale and thresholded with the same rules, so the reward table can act

only through the learned scores; at the fixed cut-off $\hat{p} \geq 1/6$ (rule D0) the symmetric-reward variant is not worse either (cost 0.825 vs. 0.905), although the effective position of that cut-off on $P(y = 1 \mid s)$ differs between the variants (Section III-D).

Varying the reward's regret ratio from 1:1 to 10:1 with E-DDQN's configurations fixed moves the greedy policy along the trade-off as Eq. (3) predicts (recall 52.4 % at 1:1, 77.2 % at 5:1, 80.7 % at 10:1), but once the threshold is chosen on validation data all ratios end at similar costs (0.629–0.673; within one seed SD, untested; Supplementary Sec. S9.6): the ratio sets the default operating point, not the discrimination.

G. TELEMETRY-PERTURBATION STRESS TEST (RQ3)

Under the whole-session perturbations, 87 % of the model–scenario ROC-AUC values change by less than 0.03 and none by more than 0.089 (Supplementary Sec. S9.7). This feature-space stress test gives no evidence of robustness: 29 of 35 selected configurations use exposure features, which the perturbations barely touch, and a shift applied to every session alike need not reorder scores. At fixed thresholds the adaptive adversary lowers recall on event seconds by 4–19 percentage points and raises false-positive rates to 94.5–100 % (Supplementary Sec. S4.3). SHAP attributions of E-DDQN's ΔQ are dominated by cum_rot, elapsed_s, speed_mean_sofar, CSPQ VR experience: some, cum_acc_spikes (Supplementary Sec. S9.8): exposure and questionnaire inputs rather than instantaneous telemetry, consistent with the exposure-time baselines; since the primary E-DDQN ranks near chance, they describe what it uses, not a validated signal.

V. DISCUSSION

A. WHAT THE RESULTS SUPPORT

Evaluation. On these data the split protocol, not the model, decides whether scores look near-perfect; accuracy figures on gameplay cybersickness data should be read together with the split protocol.

RQ1. On logged data the asymmetric reward encodes a cost-sensitive operating point: under the natural class mix the greedy policy is the 5:1 Bayes rule, and class-balanced replay shifts it. Empirically the ratio moved the default operating point but not the discrimination, and replacing the asymmetric reward with the symmetric one lowered rather than raised E-DDQN's cost as a point estimate (Section IV-F). The evidence therefore does not support a benefit of the asymmetric reward beyond what cost weighting and a validation-chosen threshold provide.

RQ2. E-DDQN did not have a significantly lower 5:1 cost than the standard DQN with any of three search draws, and the two agents' ranking reversed between draws, so neither is consistently better. With its primary draw its seed-ensemble cost was higher than the random forest's, significantly under the primary procedure but not under the sensitivity analysis, so this is not firmly established. No primary model ranked seconds significantly better than elapsed time, and no model meets a safety-prioritized target.

RQ3. The small effect of bounded perturbations on ranking reflects which inputs the selected configurations use; at fixed thresholds an adaptive adversary still changes decisions. These feature-space stress tests say nothing about real rendering attacks.

B. IS E-DDQN REINFORCEMENT LEARNING?

With logged data the alarm cannot influence later telemetry, so the problem is a contextual bandit. E-DDQN remains a valid value-based learner of a cost-sensitive policy, but bootstrapping cannot improve the optimal policy and, under bandit feedback, the agent observes the reward of one action although the labels reveal both—which is why full-information replay is the principled way to use the labels. Any DQN classifier trained on a fixed dataset whose episodes end only with the data must therefore gain through estimation effects and should be compared with cost-weighted models at the same operating point; error-triggered termination [11], [12] only adds a γ-dependent penalty on that error. Adaptive sequential control would require an alarm that triggers a mitigation whose effect on later telemetry and symptoms is measured, which, to our knowledge, no public cybersickness dataset provides.

C. THE SAFETY OBJECTIVE

A 5:1 cost makes an alarm cheaper whenever $P(y = 1 \mid s) \geq 1/6$. With a prevalence of 0.44 and weakly informative inputs almost every second qualifies, so every cost-optimal policy is close to always alarming: high recall, but 82–97 % false alarms. High recall alone is therefore not evidence of a useful monitor. Judged on cost, threshold-free ranking and cost curves across ratios, near-complete detection with a controlled false-alarm rate is not achieved on these data. Once sessions and groups are not shared, no primary model ranked seconds significantly better than session duration, which distinguishes individual susceptibility only coarsely; questionnaire indicators, which could in principle capture susceptibility, appeared not to transfer across the 21 groups (exploratory; Section IV-E).

D. LIMITATIONS

(i) One public dataset with 35 sessions, 21 groups and 26 onsets: intervals are wide, fold-to-fold variation is large, and inference with so few clusters is approximate (E-DDQN vs. random forest: Holm-adjusted $p = 0.006$ by bootstrap, 0.083 by sign-flip). (ii) Participant identity cannot be verified: linkage groups are reconstructed from questionnaire profiles and session timing; the checks of Supplementary Sec. S8 establish session- and group-disjoint evaluation, not participant independence. (iii) The label is a coarse, held self-report with undocumented elicitation, and the binary threshold merges slight with severe discomfort, which is almost absent from Flight. (iv) All analyses beyond the

TABLE 5. Leave-one-component-out ablation of E-DDQN (selected configuration per outer fold held fixed; rule D1; mean over 5 seeds, cost ± SD). The no-warm-up variant changes only the one fold whose selected configuration used warm-up.

Variant	Recall	Spec.	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC
Full E-DDQN (selected)	93.4	7.4	0.663±0.048	0.538	0.483
full-information replay toggled	93.7	5.4	0.667±0.106	0.563	0.500
symmetric ±1 reward instead of asymmetric	94.8	8.6	0.625±0.038	0.572	0.517
uniform instead of class-balanced replay	94.3	4.3	0.660±0.045	0.554	0.486
no double estimator (max-operator target)	93.1	5.3	0.681±0.086	0.543	0.491
no dueling head	94.6	5.8	0.644±0.059	0.559	0.494
hard target copy every 10 updates	94.9	6.5	0.634±0.043	0.529	0.472
no warm-up	93.8	7.1	0.656±0.049	0.547	0.497
$\gamma = 0$ (contextual bandit)	94.4	6.4	0.645±0.070	0.576	0.510

frozen protocol are post hoc; Section III-G and Supplementary Sec. S5 state which were committed before being run (the exposure-time baselines, Holm adjustment of the E-DDQN comparisons, always-alarm comparison, grouping sensitivity, cost curves and the search-draw experiment) and which were added later, some after a pre-submission review when all results were known. (v) Perturbations are synthetic and feature-space, their physiological effect is unknown, and the adversary is weaker than optimization-based attacks. (vi) The search budget (16 configurations; 12 for logistic regression) covers a far smaller fraction of the RL spaces (about 3.0 million E-DDQN configurations) than of the random-forest space (192); only E-DDQN and the DQN were re-tuned with further draws, so the other rankings may also depend on their draws. (vii) The October-2019 sessions could not serve as an external test set because their orientation channels are constant (a logging defect). (viii) The diagnosis of E-DDQN's weak result rests on three draws and one feature-set restriction and does not isolate any single component. (ix) Thresholds are chosen on OOF scores from single inner models but applied to test scores averaged over three inner models. Averaging narrowed the score spread by a similar factor for every model (median SD ratio 0.85 for E-DDQN, 0.85–0.93 for the others). In a post-hoc check, applying the unchanged thresholds to single-model test scores instead changed the seed-mean 5:1 cost by +0.045 to +0.067 for the neural models (E-DDQN +0.054) and by −0.009 to +0.014 for the others, and left the cost ranking of the seven models unchanged (Supplementary Sec. S9.10). The scale difference therefore does not explain E-DDQN's weak result. (x) For models trained on a balanced class mix and for the standard DQN, the untuned rule D0 is the 5:1 Bayes cut-off only nominally; D1 and the ranking metrics are affected only marginally (Section III-D).

E. IMPLICATIONS

Results on session data should be reported with participant- or session-grouped splits, trivial policies and an exposure-time baseline. RL detectors compared under small search budgets should be evaluated over several random draws; here E-DDQN's within-fold ROC-AUC spanned 0.175 across three draws. An asymmetric reward should be treated as a cost-sensitive operating point and compared with cost-weighted models at matched recall or cost. Robustness claims should state which inputs a detector uses and evaluate decisions at fixed thresholds. Progress is likely to require physiological or eye-tracking signals, verified participant identifiers and, for adversarial claims, manipulations applied to participants with measured effects.

VI. CONCLUSION

We asked whether an asymmetric-reward dueling Double DQN (E-DDQN) improves safety-prioritized cybersickness detection in VR; on the public UFFCSData gameplay dataset we found no evidence that it does. On logged data the problem is a contextual bandit, and the 5:1 reward induces exactly the Bayes threshold of 5:1 cost-sensitive classification. Under a nested, session-linkage-grouped protocol fixed before testing, E-DDQN reached a 5:1 cost of 0.663 (mean over seeds; seed ensemble 0.726) and ROC-AUC 0.538, and did not have a significantly lower cost than a standard DQN (0.584; ensemble 0.536) in any of three search draws. Its seed-ensemble cost was higher than a cost-weighted random forest's (0.491; ensemble 0.485; difference +0.242, 95 % CI +0.052 to +0.528), significantly under the primary group bootstrap (Holm-adjusted $p = 0.006$) but not under the sign-flip sensitivity analysis (Holm-adjusted $p = 0.083$), so this difference is not firmly established. Across draws E-DDQN's within-fold ROC-AUC ranged from 0.539 to 0.714, which suggests that search-draw sensitivity contributes to its weak primary result without isolating a single cause. A post-hoc logistic regression on elapsed time alone reached ROC-AUC 0.704, no primary model ranked seconds significantly better, and no model combined high recall with a controlled false-alarm rate or was shown, with linkage groups as units, to improve on always alarming, which reflects limited evidence rather than demonstrated equivalence. Row-level splits of the same data give near-perfect scores, so grouped evaluation was essential. Bounded perturbations, a secondary feature-space stress test that does not evaluate real attacks, changed ROC-AUC by at most 0.089. RL detectors for this task should be compared with cost-weighted and exposure-time baselines,

over several search draws, under grouped evaluation.

DATA AND CODE AVAILABILITY

The data are the publicly available UFFCSData release [18] (AGPL-3.0). All code, the evaluation protocol (`configs/protocol.json`, committed before any model was trained on the evaluation sessions), the specification of the post-hoc analyses (`configs/posthoc.json`; which analyses were committed before being run and which were added later is stated in Supplementary Sec. S5), the split tables, the per-sample predictions of every model and the scripts that produce every number, table and figure of this paper are archived at [DOI to be added on deposit]. A single script regenerates everything from the raw XML files. Seeds are fixed for data splitting (20261012), hyperparameter search (777; 801–802 for the post-hoc baselines; six configuration seeds between 9120 and 9202 for the search-draw analysis) and final training (1000–1004); CPU execution with one thread per process was used, and minor floating-point differences across hardware are possible.

ETHICS STATEMENT

This study is a secondary analysis of the publicly released UFFCSData dataset [18], whose session files contain no names or contact details of participants. No new data were collected from human participants, and no attempt was made to identify individuals; the session-linkage groups (Section III-B) are used only to keep sessions that may belong to the same person in the same data partition. [Add your institution's ruling if it requires a formal exemption for secondary analysis of public data, and the ethics approval of the original data collection if the journal asks for it.]

AUTHOR CONTRIBUTIONS

[To be completed by the authors with CRediT roles]: R. Darabee: [roles]; M. Muniandy: [roles]; O. Shindi: [roles]; C. W. Khuen: [roles]; E. H. Sumiea: [roles]. All authors read and approved the final manuscript.

USE OF GENERATIVE AI

The authors used Claude, an AI assistant developed by Anthropic ([version and period of use]), to help re-implement and run the analysis pipeline (data parsing, evaluation protocol, statistical analysis and figures), to check numerical consistency and bibliographic references, and to draft and edit the manuscript text. The authors reviewed and verified the code, results and text and take full responsibility for the content of this article. [Confirm that this describes your use, and adapt it to the target journal's policy.]

ACKNOWLEDGMENT

The authors thank Universiti Teknologi PETRONAS for institutional and computational support, and the authors of the UFFCSData dataset for making their data public.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

FUNDING

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

REFERENCES

[1] P. Caserman, A. Garcia-Agundez, A. Gámez Zerban, and S. Göbel, "Cybersickness in current-generation virtual reality head-mounted displays: Systematic review and outlook," Virtual Reality, vol. 25, no. 4, pp. 1153–1170, 2021.

[2] W.-J. Tseng, E. Bonnail, M. McGill, M. Khamis, E. Lecolinet, S. Huron, and J. Gugenheimer, "The dark side of perceptual manipulations in virtual reality," in Proc. CHI Conf. Human Factors in Computing Systems, 2022.

[3] B. Odeleye, G. Loukas, R. Heartfield, and F. Spyridonis, "Detecting framerate-oriented cyber attacks on user experience in virtual reality," in Proc. 1st Int. Workshop Security for XR and XR for Security (VR4Sec), held with SOUPS 2021, 2021, pp. 1–5.

[4] R. K. Kundu, M. Denton, G. Mongalo, P. Calyam, and K. A. Hoque, "Securing virtual reality experiences: Unveiling and tackling cybersickness attacks with explainable AI," IEEE Trans. Dependable Secure Comput., vol. 22, no. 6, pp. 6040–6057, 2025, arXiv:2503.13419.

[5] M. S. Dennison, A. Z. Wisti, and M. D'Zmura, "Use of physiological signals to predict cybersickness," Displays, vol. 44, pp. 42–52, 2016.

[6] R. Islam, Y. Lee, M. Jaloli, I. Muhammad, D. Zhu, P. Rad, Y. Huang, and J. Quarles, "Automatic detection and prediction of cybersickness severity using deep neural networks from user's physiological signals," in Proc. IEEE Int. Symp. Mixed and Augmented Reality (ISMAR), 2020, pp. 400–411.

[7] R. Islam, K. Desai, and J. Quarles, "Cybersickness prediction from integrated HMD's sensors: A multimodal deep fusion approach using eye-tracking and head-tracking data," in Proc. IEEE Int. Symp. Mixed and Augmented Reality (ISMAR), 2021, pp. 31–40, arXiv:2108.06437.

[8] M. Liu, B. Yang, M. Xu, P. Zan, L. Chen, and X. Xia, "Exploring quantitative assessment of cybersickness in virtual reality using EEG signals and a CNN-ECA-LSTM network," Displays, vol. 81, p. 102602, 2024.

[9] M. Yalcin, A. Halbig, M. Fischbach, and M. E. Latoschik, "Automatic cybersickness detection by deep learning of augmented physiological data from off-the-shelf consumer-grade sensors," Frontiers in Virtual Reality, vol. 5, p. 1364207, 2024.

[10] D. Jeong and K. Han, "PRECYSE: Predicting cybersickness using transformer for multimodal time-series sensor data," Proc. ACM Interact. Mob. Wearable Ubiquitous Technol., vol. 8, no. 2, pp. 42:1–42:24, 2024.

[11] E. Lin, Q. Chen, and X. Qi, "Deep reinforcement learning for imbalanced classification," Applied Intelligence, vol. 50, pp. 2488–2502, 2020.

[12] J. Yang, R. El-Bouri, O. O'Donoghue, A. S. Lachapelle, A. A. S. Soltan, D. W. Eyre, L. Lu, and D. A. Clifton, "Deep reinforcement learning for multi-class imbalanced training: Applications in healthcare," Mach. Learn., vol. 113, pp. 2655–2674, 2024.

[13] H. van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," in Proc. AAAI Conf. Artificial Intelligence, 2016, pp. 2094–2100.

[14] Z. Wang, T. Schaul, M. Hessel, H. van Hasselt, M. Lanctot, and N. de Freitas, "Dueling network architectures for deep reinforcement learning," in Proc. Int. Conf. Machine Learning (ICML), 2016, pp. 1995–2003.

[15] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, p. 100804, 2023.

[16] R. K. Kundu, R. Islam, P. Calyam, and K. A. Hoque, "TruVR: Trustworthy cybersickness detection using explainable machine learning," in Proc. IEEE Int. Symp. Mixed and Augmented Reality (ISMAR), 2022, pp. 777–786.

[17] C. Elkan, "The foundations of cost-sensitive learning," in Proc. Int. Joint Conf. Artificial Intelligence (IJCAI), 2001, pp. 973–978.

[18] T. Porcino et al., "UFFCSData: Raw and selected datasets with user tests reporting the level of discomfort during virtual experiments," https://github.com/tmp1986/UFFCSData, 2019, AGPL-3.0.

[19] M. Dudík, D. Hsu, S. Kale, N. Karampatziakis, J. Langford, L. Reyzin, and T. Zhang, "Efficient optimal learning for contextual bandits," in Proc. Conf. Uncertainty in Artificial Intelligence (UAI), 2011, arXiv:1106.2369.

[20] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal, "Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness," Int. J. Aviation Psychology, vol. 3, no. 3, pp. 203–220, 1993.
[21] R. K. Kundu, R. Islam, J. Quarles, and K. A. Hoque, "LiteVR: Interpretable and lightweight cybersickness detection using explainable AI," in Proc. IEEE Conf. Virtual Reality and 3D User Interfaces (VR), 2023, pp. 609–619.
[22] R. K. Kundu, O. Y. Elsaid, P. Calyam, and K. A. Hoque, "VR-LENS: Super learning-based cybersickness detection and explainable AI-guided deployment in virtual reality," in Proc. 28th Int. Conf. Intelligent User Interfaces (IUI), 2023, pp. 819–834.
[23] A. H. X. Yang, N. Kasabov, and Y. O. Cakmak, "Machine learning methods for the study of cybersickness: A systematic review," Brain Informatics, vol. 9, p. 24, 2022.
[24] T. Porcino, D. Trevisan, and E. Clua, "An experimental methodology to capture user and gameplay data tied to cybersickness," in Proc. Workshop on Extended Reality in Games (XR in Games), held with the ACM Int. Conf. Interactive Media Experiences (IMX), 2021.
[25] T. Porcino, E. O. Rodrigues, F. Bernardini, D. Trevisan, and E. Clua, "Identifying cybersickness causes in virtual reality games using symbolic machine learning algorithms," Entertainment Computing, vol. 41, 2022, art. no. 100473.
[26] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," ACM Trans. Knowl. Discov. Data, vol. 6, no. 4, p. 15, 2012.
[27] D. R. Roberts et al., "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," Ecography, vol. 40, no. 8, pp. 913–929, 2017.
[28] B. Odeleye, G. Loukas, R. Heartfield, G. Sakellari, E. Panaousis, and F. Spyridonis, "Virtually secure: A taxonomic assessment of cybersecurity challenges in virtual reality environments," Computers & Security, vol. 124, p. 102951, 2023.
[29] A. Giaretta, "Security and privacy in virtual reality: A literature survey," Virtual Reality, vol. 29, p. 10, 2025.
[30] K. Cheng, J. F. Tian, T. Kohno, and F. Roesner, "Exploring user reactions and mental models towards perceptual manipulation attacks in mixed reality," in Proc. 32nd USENIX Security Symp. USENIX Association, 2023. [Online]. Available: https://www.usenix.org/conference/usenixsecurity23/presentation/cheng-kaiming
[31] I. Ahmed, R. K. Kundu, and K. A. Hoque, "Adversarial VR: An open-source testbed for evaluating adversarial robustness of VR cybersickness detection and mitigation," in Proc. IEEE Int. Symp. Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), 2025.
[32] V. Mnih et al., "Human-level control through deep reinforcement learning," Nature, vol. 518, no. 7540, pp. 529–533, 2015.
[33] J. García and F. Fernández, "A comprehensive survey on safe reinforcement learning," J. Machine Learning Research, vol. 16, no. 42, pp. 1437–1480, 2015.
[34] T. Porcino, D. Trevisan, and E. Clua, "DEMO: Using gameplay data to classify cybersickness level in virtual environments," in Anais Estendidos do Simpósio de Realidade Virtual e Aumentada (SVR). SBC, 2019. [Online]. Available: https://sol.sbc.org.br/index.php/svr_estendido/article/view/8462
[35] T. Lattimore and C. Szepesvári, Bandit Algorithms. Cambridge University Press, 2020.
[36] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in Proc. Int. Conf. Machine Learning (ICML), 2015, pp. 448–456.
[37] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," J. Machine Learning Research, vol. 15, pp. 1929–1958, 2014.
[38] P. J. Huber, "Robust estimation of a location parameter," Ann. Math. Statist., vol. 35, no. 1, pp. 73–101, 1964.
[39] R. Girshick, "Fast R-CNN," in Proc. IEEE Int. Conf. Computer Vision (ICCV), 2015, pp. 1440–1448.
[40] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in Proc. Int. Conf. Learning Representations (ICLR), 2015.
[41] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in Proc. Int. Conf. Learning Representations (ICLR), 2017.
[42] T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," in Proc. Int. Conf. Learning Representations (ICLR), 2016.
[43] L.-J. Lin, "Self-improving reactive agents based on reinforcement learning, planning and teaching," Machine Learning, vol. 8, no. 3–4, pp. 293–321, 1992.
[44] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
[45] R. S. Kennedy, K. M. Stanney, and W. P. Dunlap, "Duration and exposure to virtual environments: Sickness curves during and across sessions," Presence: Teleoperators and Virtual Environments, vol. 9, no. 5, pp. 463–472, 2000.
[46] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in Advances in Neural Information Processing Systems (NeurIPS), vol. 32, 2019.
[47] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," BMC Genomics, vol. 21, p. 6, 2020.
[48] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLoS ONE, vol. 10, no. 3, p. e0118432, 2015.
[49] S. Holm, "A simple sequentially rejective multiple test procedure," Scand. J. Statist., vol. 6, no. 2, pp. 65–70, 1979.
[50] A. C. Cameron, J. B. Gelbach, and D. L. Miller, "Bootstrap-based improvements for inference with clustered errors," Review of Economics and Statistics, vol. 90, no. 3, pp. 414–427, 2008.
[51] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," Psychometrika, vol. 12, no. 2, pp. 153–157, 1947.
[52] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," Neural Computation, vol. 10, no. 7, pp. 1895–1923, 1998.
[53] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems (NeurIPS), 2017, pp. 4765–4774.
[54] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," in Proc. Int. Conf. Machine Learning (ICML), 2017, pp. 3145–3153.
[55] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," BMC Bioinformatics, vol. 7, p. 91, 2006.

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